Politecnico di Milano
Gaia Biasolo, Matteo Giuseppe Cigada, Federico D’Agostini, Simone D’Incà

Lab 09.2: Capgemini Seminar 2

05/12/2025

Abstract

This report details the methodology and application of **Design Failure Mode & Effect Analysis (D-FMEA)** as a cornerstone of preventive reliability. The primary goal is to achieve zero defects by identifying potential failures during the design phase, thereby minimizing "non-quality costs" associated with new product introductions.

Core Methodology

The report outlines a structured seven-step process designed to improve information flow and facilitate critical design discussions:

- **Preparation & Analysis**: initial steps involve planning with cross-functional teams, structure analysis using block diagrams, and function analysis via correlation matrices
- **Risk Evaluation**: failure analysis identifies causes (using the "5 Whys") and effects, assigning a Risk Priority Number (RPN) based on Severity, Occurrence, and Detection
- **Optimization**: a corrective action plan is implemented to reduce RPN values; notably, while occurrence and detection can be improved, the Severity score remains constant as it depends on the failure's effect on the customer

Contents

1. The Design FMEA

1.1 Preventive Reliability

- **Definition: it is the set of activities that allow to obtain the quality expected by the customer and keep it in its lifecycle.**
- **Goal: zero defects/issues**
- Two methods used
  - **FMEA** (during the design phase)
  - **Problem Solving** (during experimental, production and working phases)
- Output: database of “lessons learned”

1.1.1 Preventive reliability process and cycle

- preventive reliability activities occur during the Specifications, System, and Design phases
- Concept → design → validation →industrial → production → service → customer
  - issues found in Industrial, production, or service phases trigger problem solving
  - problem solving generates "Lesson Learned" data, which feeds back into the concept phase

1.1.2 Reliability plan and importance of preventive reliability

- action is split into two levels**: Root Cause Analysis** for **carry-over contents** and **Preventive Reliability** for **new contents**
- cost for current production product: only carry over costs
- cost for new product: carry over costs for the previous development, innovation costs and carry over costs for the new development (the sum of the three is addressed as “non quality cost”
- cost scenarios for a new product based on the analysis method used:
  - without Preventive Analysis: high non quality costs
  - focus on New Content: moderate non quality costs
  - global Preventive Analysis: lowest non quality costs
- ⟹ **preventive reliability is needed in the design phase to keep the costs low in the introduction of a new product**

1.2 D-FMEA Methodology

1.2.1 Definition

- **Design - Failure Mode & Effect Analysis is a preventive tool**
- **it identifies potential failure modes/effects, potential root causes, and critical design characteristics (input for Production-FMEA)**
- **it analyses the validation plan to detect failures before SOP and defines a corrective action plan to reduce risk**
- steps:
    1. Planning & Preparation
    2. Structure Analysis
    3. Function Analysis
    4. Failure Analysis
    5. Risk Analysis
    6. Optimization
    7. Results Documentation

1.2.2 Step 1: Planning & Preparation

- a cross-functional team is essential, including Engineering, Testing, Manufacturing, Quality, Production, and Service
- **objective:** improve information flow and facilitate critical discussion to identify failures and design parameters that affect functionality
- The methodology relies on four key inputs
    - **engineering:** design criteria and drawing quality analysis
    - **manufacturing:** design for manufacturing and plant issue analysis
    - **quality:** field issues and Lesson Learned implementation. Goal: avoid repeating the same mistakes
    - **testing:** experimental validation issues

- Workflow:
  - DFMEA
  - drawing analysis
  - improvement proposals

1.2.3 Step 2: Structure analysis

- block diagram is used
  - it defines elements and highlight interactions (interfaces)
  - a dashed line is used to define the scope of analysis (inside perimeter = considered in FMEA)
  - Arrows indicate the connections
    - Internal/external
    - Physical/non physical

1.2.4 Step 3: Function Analysis

- Correlation matrix is used
  - It maps "Functions" (rows) against "Elements & Interfaces" (columns)
  - It links functions with elements and interfaces
  - It ensures that no interaction is neglected
  - Critical interactions can also be identified
  - The intersection where a specific element provides a specific function is marked with an "X"

1.2.5 Step 4: Failure Analysis

- Risk matrix is used
  - It allows to five priority to risks
  - Structure from left to right: Object → Effects → Cause → Critical characteristics → Criteria Evaluation → Risks
- **Object**
  - Identifies the Item (Element/Interface) and its Elementary Function to be analyzed
    - the item is the component indicated in the correlation matrix and block diagram

        - the elementary function is the job that must be carried out by the item

- **Effects**
    - Describe what is perceived by the "Customer" (Final User, Plant, or Service) if a function is not guaranteed
    - A **<u>severity</u>** value 1-10 is assigned to each effect of failure
        - Severity 1,2,3 → low effect from customer point of view, low impact
        - Severity 8,9,10 → safety of the user/homologation is compromised
- **Cause**
    - define *how* the effect happens and lists all failure modes that can cause the identified effects
    - **It is the opposite of the elementary function**
    - 5 whys method → climbing up the series of causes to find the root one
    - root causes are defined as corrections that need to be applied in the drawings to avoid the failure
- **Critical characteristics**
    - Goal: identify geometric parameters where variation degrades performance, even if technically robust
    - This step is needed to guide the industrialization and invest budget where needed → improve the control plan
    - The class is a number associated with the root cause identified in the drawings
- **Criteria evaluation**
    - verifies design criteria against standards and virtual validation
    - description of how the verification is going to be performed to test the failure mode
    - **<u>Occurrence</u>** evaluates the robustness of the criteria (low occurrence = the criteria used to verify the failure is very reliable es. standard testing procedure)
    - list of experimental tests for the validation plan

- **<u>Detection</u>** indicates how well does the test verify if failure takes place or not (standardized tests and worst case scenario tests have the best detection rates)

- **Risks**
    - RPN is computed as the product of severity, occurrence and detection

1.2.6 Step 5: Risk Analysis

- High RPN = high risk of failure
- Low RPN = low risk of failure/robust design
- Thresholds are set by companies
- Continuous improvement approach: improvements can be made even if scores are below the threshold
- Action Priority Table can be used as an alternative to RPN
    - Based on combinations/interactions of Severity, Occurrence and Detection scores
    - Final output is a level of priority: Low, Medium, High

1.2.7 Step 6: Optimization

- It is the corrective action plan
    - Lists Recommended Action, Responsibility, Due Date, and the "Result" (new O, D, RPN values) after implementation of the correction

- Severity does not change with the implementation of the action, it depends only on the function and on the effect from the customer point of view

1.2.8 Step 7: Results Documentation

- indicators for meeting schedules, analysis progress (Open vs. Closed points), and corrective action status
- graphs show the progress of "To Be Done," "Working," and "Completed" tasks over time

2. DFMEA Case study

- **Client & Subject:** The study was conducted for IVECO (Automotive) focusing on the **Front Bumper** system of the Stralis MY24 truck
- **Objective:** To analyze critical aspects of design, functionality, ergonomics, and installation to suggest corrective actions and improve final product quality
- **Decomposition:** The bumper was broken down into specific sub-components for analysis, including the Upper/Lower Deflectors, Foldable Step, Antimosquitos Grid, Headlamp Covers, and various reinforcement brackets
- **Block Diagram:** A boundary diagram was created to map the physical and functional connections between these components

2.1 Radar Interference (ADAS System)

- **The Issue:** interaction between the Foldable Step and the ADAS Frontal Radar
- **Failure Mode:** the foldable step physically obstructed the radar's field of view (signal transmission cone), leading to ADAS malfunction
- **Initial Risk:** high RPN of **126** (Severity: 9, Occurrence: 7)
- **Corrective Action:** performed a geometric stack-up analysis to verify alignment against supplier prescriptions.
- **Result:** the risk was mitigated, reducing the RPN to **72** (Occurrence dropped from 7 to 4)

2.2 Corrosion & Fixing (Step Cover)

- **The Issue:** fixing mechanism between the Step Plastic Covers and the Foldable Step
- **Failure Mode:** risk of component detachment due to screw breakage caused by corrosion (inadequate design for environmental resistance)
- **Initial Risk:** high RPN of **112** (Severity: 7, Occurrence: 8)
- **Corrective Action:** the screw specification was changed to a type compatible with a 1000-hour corrosion resistance target (up from the previous 720-hour standard)
- **Result:** the risk was significantly lowered, reducing the RPN to **42** (Occurrence dropped from 8 to 3)